Random and Adversarial Positional-Parameter Robustness Decouple Across Positional-Encoding Families

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Abstract

Positional encodings (PEs) shape how Vision Transformers convert parameter perturbations into functional failure. We compare Learned, Sinusoidal, RoPE, and ALiBi under random and adversarial perturbations using relative RMS displacement of pre-softmax attention logits (ρ_{rel}) as a common scale. Across 72 canonical checkpoints spanning ViT-B/16 on CIFAR-100 and ImageNet-100 and ViT-S/16 on ImageNet-100, the mapping from displacement to classification damage depended on PE family and architecture. The gap was positive throughout, but its magnitude and adversarial ordering varied across dataset and architecture. An 18-checkpoint ViT-S/16 AMP cohort enabled an AMP-matched comparison with ViT-B/16. ViT-S/16 had lower adversarial nAUC for Learned, Sinusoidal, and RoPE but nearly unchanged noise nAUC; the architecture effect varied by family (exact Friedman $p = 0.02894$) and reversed the Learned–RoPE ordering. Within ViT-S/16, family rankings were preserved across AMP and full-FP32 training, but formal ρ_{50} equivalence held only for Sinusoidal. The ViT-S/16 AMP–FP16 ALiBi cohort did not yield valid checkpoints, whereas the recipe was stable for ViT-B/16 AMP and ViT-S/16 full-FP32, indicating an architecture-dependent trainability boundary. At a single ViT-S/16 Sinusoidal checkpoint (seed 42) and budget 0.0095, direct-displacement optimization produced larger displacement ($\rho_{\mathrm{rel}} = 0.1021$ versus 0.0938) yet retained 76.70% versus 7.36% accuracy under task-loss

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optimization. An all-restart audit ruled out restart selection as the cause of the Sinusoidal task-loss frontier; layerwise profiles showed objective-dependent redistribution. These results show that architecture, objective, training regime, random stability, and displacement require separate evaluation.

Keywords: Vision Transformer, positional encoding, adversarial parameter perturbation, random parameter noise, attention robustness, local geometry

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1. Introduction

Positional information is indispensable to Vision Transformers (ViTs), whose self-attention operation is otherwise permutation equivariant with respect to the input-token sequence. Learned absolute embeddings, deterministic sinusoidal tables, rotary positional encoding (RoPE), and attention with linear biases (ALiBi) serve a common functional purpose but enter the computation graph through fundamentally different operations. Learned and sinusoidal encodings are added to token representations, RoPE changes query–key geometry through position-dependent rotations, and ALiBi contributes a position-dependent bias directly to attention logits.

This diversity makes native parameter-space robustness comparisons non-commensurate. Equal parameter norms across these mechanisms need not produce equal internal changes because the spaces differ in dimensionality, sharing structure, scale, and mapping into attention. We therefore measure the achieved relative root-mean-square displacement of pre-softmax attention logits, denoted ρ_{rel}, and analyse accuracy as a function of that achieved displacement rather than native perturbation budget alone.

A second distinction concerns perturbation direction. Random parameter noise samples typical directions from a prescribed distribution, whereas an adversarial perturbation searches for directions that increase task loss. Equal ρ_{rel} controls aggregate displacement magnitude but not its orientation, layer allocation, head or token structure, or alignment with task-critical directions. Consequently, random robustness can fail as a proxy for task-aligned robustness even after achieved attention-logit displacement magnitude is matched.

We evaluate Learned, Sinusoidal, RoPE, and ALiBi in a canonical AMP-trained ViT-B/16 cohort on CIFAR-100 and ImageNet-100 and in an independent full-FP32 ViT-S/16 ImageNet-100 replication. Each positional-encoding family uses six trained seeds. A separate completed ViT-S/16 AMP cohort for

Learned, Sinusoidal, and RoPE enables an AMP-matched cross-architecture comparison. Within an architecture, random and adversarial curves are compared on the same checkpoints and seed identities. Across architectures, equal seed labels provide seed-aligned descriptive contrasts but not strict training pairs because initialization dimensionality and optimization trajectories differ. We organize the three available contrasts as a factorial decomposition: the four-family canonical cross-cohort comparison jointly changes architecture and training precision; the within-ViT-S/16 training-regime comparison holds architecture fixed and varies training precision; and the AMP-matched cross-architecture comparison holds AMP training fixed and varies architecture for the three shared families. This structure separates system-level replication, precision transfer, and backbone effects without treating equal seed labels as strict training pairs.

The ViT-B/16 results show a positive noise-minus-adversarial robustness gap in every seed, dataset, and family, together with dataset- and family-dependent gap magnitudes and a wider-range CIFAR-100 ordering reversal. The ViT-S/16 replication preserves the random-noise extremes—RoPE highest and ALiBi lowest—but changes the adversarial ordering substantially. In particular, Learned PE is strongest adversarially for ViT-B/16 but falls below ALiBi and RoPE for ViT-S/16, while ALiBi changes from the weakest ViT-B/16 adversarial family to the strongest ViT-S/16 family over the locked architecture-common support.

We additionally audit the interpretation of Sinusoidal robustness. An all-restart task-loss audit records every restart rather than only the loss-selected point, and a separate direct-ρ optimization asks how far attention logits can be moved when displacement itself is optimized. These experiments show that the observed boundary is a *task-loss frontier*, not a structural upper bound: direct-ρ optimization reaches much larger displacement, while perturbations with similar global displacement can have radically different classification consequences. A layerwise decomposition shows that direct-ρ and task-loss directions allocate displacement differently across Transformer blocks.

The principal contributions are:

- an achieved attention-logit displacement scale for comparing perturbations across structurally different positional mechanisms;
- a 72-checkpoint canonical evaluation covering AMP-trained ViT-B/16 on two datasets and a full-FP32 ViT-S/16 ImageNet-100 replication, plus an 18-checkpoint ViT-S/16 AMP auxiliary cohort;

- a factorial architecture–precision decomposition with stable random-noise extremes but PE-dependent adversarial ranking, mean backbone effects, and seed dispersion;
- an all-restart and direct-displacement analysis establishing an objective-specific Sinusoidal task-loss frontier and separating global displacement magnitude from functional damage; and
- local and layerwise analyses that distinguish parameter-to-attention amplification, task-selective harmfulness, and inter-layer displacement allocation.

2. Related Work

2.1. Positional encoding in Transformers and Vision Transformers

Self-attention does not encode token order by itself, motivating explicit positional information in the original Transformer (Vaswani et al., 2017). Vision Transformer adopted learned absolute embeddings for image-patch sequences (Dosovitskiy et al., 2021); RoPE injects relative structure through position-dependent query–key rotations (Su et al., 2024); and ALiBi adds head-dependent distance penalties directly to pre-softmax attention scores (Press et al., 2022). Dufter et al. (2022) surveys these mechanisms in a unified notation, emphasizing that positional information can enter a Transformer through structurally different operations.

Vision-specific alternatives reinforce this heterogeneity. iRPE develops directional two-dimensional relative encodings (Wu et al., 2021b); conditional positional encoding derives position information from the local token neighbourhood (Chu et al., 2023); Swin uses learned relative biases inside shifted local windows (Liu et al., 2021); and CvT shows that convolutional token embedding and projection can partly replace explicit positional embeddings (Wu et al., 2021a). These studies mainly evaluate accuracy, efficiency, and resolution or length generalization. We instead ask a post-training robustness question: whether structurally different PE mechanisms retain the same ordering under random and task-aligned perturbations when achieved attention-logit displacement is matched.

2.2. Input-space adversarial robustness of Vision Transformers

Most adversarial-robustness work perturbs the input. Gradient-based and projected-gradient attacks established the standard norm-constrained threat model (Goodfellow et al., 2015; Madry et al., 2018). ViT studies then examined white- and black-box attacks, transferability, corruptions, occlusions, spatial permutations, and architecture-controlled comparisons (Mahmood et al., 2021; Naseer et al., 2021; Bai et al., 2021). Attack design itself can change apparent rankings: Patch-Fool adapts the attack to patch and attention structure (Fu et al., 2022), while adversarial training and patch-focused studies show sensitivity to pretraining, optimization, and perturbation allocation (Mo et al., 2022; Liu et al., 2023).

Most directly related to our attack surface, Adversarial Positional Embeddings (AdPE) perturbs absolute or relative positional information during masked-image pretraining, using embedding- and coordinate-based forms to harden the pretext task (Wang et al., 2023). Its goal is representation learning rather than post-training robustness evaluation, but it establishes PE itself as an adversarially manipulable ViT component. In our setting the image is fixed, all non-positional parameters are frozen, and only the PE-specific parameter or inference-time positional state is perturbed.

2.3. Robustness to neural-network parameter perturbations

Parameter perturbations have been studied in optimization, generalization, and deployment security. Adversarial Weight Perturbation optimizes against locally harmful weight changes during adversarial training (Wu et al., 2020); bounded weight perturbations have also been analysed theoretically (Tsai et al., 2021), and random or multiplicative perturbations used as regularizers (Trinh et al., 2024). Random-neighbourhood and SAM/AWP-style worst-case objectives likewise probe different local properties (Li et al., 2024).

Post-training security work provides a qualitative precedent for our central question. Targeted bit flips can collapse accuracy while many random flips do little (Rakin et al., 2019); small parameter changes can implant backdoors (Garg et al., 2020); and adversarial parameter attacks can selectively reduce robustness without equally conspicuous clean degradation (Yu et al., 2023). These studies usually perturb broad weight collections and measure magnitude in native parameter coordinates. That is appropriate within one parameterization, but does not directly compare Learned, Sinusoidal, RoPE, and ALiBi, whose positional states differ in dimension, sharing, constraints, and computational effect. We therefore freeze all non-positional parameters and compare perturbations by the attention-logit displacement they induce.

2.4. Random and adversarial directions in local model geometry

Loss-landscape research helps interpret random versus objective-aligned directions. Sharp minima have been linked empirically to weaker generalization (Keskar et al., 2017); filter-normalized random directions reduce sensitivity to layer scale in landscape visualization (Li et al., 2018); and SAM explicitly optimizes worst-case loss in a local parameter neighbourhood (Foret et al., 2021). These results illustrate why random and task-aligned directions can probe different properties even at similar nominal magnitudes.

Native-coordinate sharpness is itself reparameterization dependent: Dinh et al. (2017) constructed functionally equivalent networks with arbitrarily different apparent sharpness. Fisher–Rao (Liang et al., 2019), normalized and relative flatness (Tsuzuku et al., 2020; Petzka et al., 2021), and ASAM (Kwon et al., 2021) address related scale or reparameterization dependence. Our comparability problem is different: a high-dimensional additive table, deterministic sinusoidal state, phase rotations, and a few shared ALiBi slopes are semantically different perturbation spaces. We therefore normalize by induced change in a common internal quantity, pre-softmax attention logits, rather than by PE-parameter scale.

We do not estimate a full Hessian, Jacobian spectrum, or global worst case. Gaussian perturbations sample typical directions in each PE-specific space, whereas task gradients and restarted PGD search objective-aligned harmful directions. The local geometry probe then separates parameter-to-attention amplification from classification damage per unit of achieved displacement.

2.5. Position of the present study

A recent theoretical complement studies trainable positional encoding in a single-layer Transformer for in-context regression and derives clean and adversarial Rademacher-complexity bounds, finding a larger vulnerability contribution from positional encoding under attack (He and Xing, 2025). Its setting is not vision, does not compare the four PE families considered here, and does not perturb a frozen post-training positional mechanism.

Our position is therefore defined by three differences. First, the threat model modifies only the positional mechanism of an otherwise frozen trained ViT, rather than pixels or patches. Second, heterogeneous PE families are compared through achieved pre-softmax attention-logit displacement rather than a shared native parameter norm. Third, the trained checkpoint/seed is the replication unit; random directions are repeated within-checkpoint measurements. This combination directly tests whether random positional-parameter stability predicts adversarial positional robustness.

3. Attention-Logit Displacement Framework

3.1. Notation

Let

$$f(x; \theta, p)$$

denote a Vision Transformer with ordinary model parameters θ and positional parameters or positional state p. The meaning of p is PE-family dependent: it denotes an additive embedding tensor for Learned and Sinusoidal PE, a rotation phase representation for RoPE, and the shared head-specific slopes used to generate attention biases for ALiBi.

For input x, let

$$Z_\ell(x; \theta, p) \in \mathbb{R}^{H\times N\times N}$$

denote the pre-softmax attention logits in Transformer block ℓ, where H is the number of heads and N is the token count. Given a perturbation δ of the positional state, define

$$\Delta Z_\ell(x; \delta) = Z_\ell(x; \theta, p + \delta) - Z_\ell(x; \theta, p).$$

For RoPE, the notation $p + \delta$ represents an additive phase perturbation followed by reconstruction of the corresponding sine and cosine tensors. This preserves the unit-circle constraint of every rotation pair. For ALiBi, the perturbed slopes are used to regenerate the complete position-dependent attention-bias tensor.

3.2. Family-specific perturbation spaces

The native perturbation variable contains only the unique positional state coordinates. Learned and Sinusoidal PE use a single additive perturbation $\delta \in \mathbb{R}^{N\times d}$ at the Transformer input. Although every Transformer block instantiates its own RoPE phase cache and applies its own copy of the same ALiBi slope vector, the analysis treats these blockwise instances as tied copies of one shared positional state. A single RoPE phase perturbation $\delta \in \mathbb{R}^{N\times d_h/2}$ and a single ALiBi slope perturbation $\delta \in \mathbb{R}^{H}$ are applied identically in all $L = 12$ blocks. The global RMS constraint is computed over these unique perturbation coordinates rather than over their repeated blockwise instances.

Perturbations act directly on the cached RoPE phase tensor. The underlying inverse-frequency buffer that generates the clean phases is held fixed and is excluded from the perturbation space. Thus, the RoPE threat model permits position-specific phase corruption but does not perturb the shared frequency law that couples clean phases across token positions. For ALiBi, the precomputed distance matrix is likewise fixed; only the shared slope vector is perturbed.

The resulting numbers of scalar perturbation degrees of freedom are Nd for Learned and Sinusoidal PE, $Nd_h/2$ for RoPE, and H for ALiBi. For ViT-B/16 on ImageNet-100 ($N = 197$, $d = 768$, $H = 12$), these are 151,296, 151,296, 6,304, and 12, respectively. For ViT-S/16 on ImageNet-100 ($d = 384$, $H = 6$), they are 75,648, 75,648, 6,304, and 6. The ViT-B/16 CIFAR-100 cohort ($N = 65$) uses 49,920, 49,920, 2,080, and 12.

3.3. Achieved attention-logit displacement

We define the absolute attention-logit displacement as

$$\rho_{\text{abs}}(\delta) = \left[\frac{1}{L} \sum_{\ell=1}^{L} \frac{1}{|\mathcal{C}| H N^2} \sum_{x \in \mathcal{C}} \| \Delta Z_\ell(x; \delta) \|_F^2 \right]^{1/2}, \tag{1}$$

where $\mathcal{C}$ is the fixed calibration set and each Transformer block receives equal weight.

Because the overall scale of clean attention logits differs across models and PE families, we normalize Eq. (1) by the clean-logit RMS:

$$s_Z = \left[\frac{1}{L} \sum_{\ell=1}^{L} \frac{1}{|\mathcal{C}| H N^2} \sum_{x \in \mathcal{C}} \| Z_\ell(x; \theta, p) \|_F^2 \right]^{1/2}. \tag{2}$$

The relative displacement is

$$\rho_{\text{rel}}(\delta) = \frac{\rho_{\text{abs}}(\delta)}{s_Z}. \tag{3}$$

The quantity ρ_{rel} is dimensionless and controls the aggregate magnitude of the induced change in pre-softmax attention logits. It does not constrain the direction of that change or its distribution across layers, heads, queries, and keys. Because the metric is defined before softmax, it also retains row-constant logit shifts even though $\text{softmax}(z + c\mathbf{1}) = \text{softmax}(z)$. Equal ρ_{rel} should therefore be interpreted as equal aggregate pre-softmax attention-logit displacement on the stated metric, not as equality of softmax-effective or attention-probability displacement across PE families.

3.4. Random and adversarial perturbation regimes

For the random regime, we sample Gaussian directions in the unique PE parameter space and normalize them using the global root-mean-square norm. A fixed sampled direction is retained across the parameter-budget grid, producing a coherent per-draw path through parameter space. Random curves are first averaged within each trained seed before any across-seed summary is computed.

For the adversarial regime, we solve

$$\max_{\delta} \frac{1}{|\mathcal{A}|} \sum_{(x,y)\in\mathcal{A}} \mathcal{L}\big(f(x;\theta, p+\delta), y\big) \quad \text{subject to} \quad \|\delta\|_{\mathrm{RMS}} \le \varepsilon, \tag{4}$$

where $\mathcal{A}$ is disjoint from the calibration and evaluation sets. The optimization is performed using projected gradient ascent with multiple random restarts.

Equation (4) is constrained in the native PE parameter space, not directly in ρ-space. Every resulting perturbation is therefore reparameterised after optimization by its achieved value of ρ_{rel} from Eq. (3).

3.5. Robustness curves and normalized AUC

For each trained model and perturbation regime, let

$$a(\rho) = \frac{\mathrm{Acc}(\rho)}{\mathrm{Acc}_{\mathrm{clean}}}$$

denote normalized classification accuracy. Normalized accuracy is not clipped to the interval $[0, 1]$. Values slightly above one can occur when a perturbed model marginally outperforms its own clean reference on the finite evaluation subset; these fluctuations are retained symmetrically rather than truncated. We include the exact clean anchor

$$(\rho, a) = (0, 1).$$

Because PGD directly maximizes classification loss and finite optimization may yield small non-monotonic fluctuations across native budgets, we construct the adversarial curve using the monotone lower envelope in increasing achieved ρ. This represents the strongest observed attack up to each displacement level.

For a common interval $[0, \rho_{\max}]$, the normalized area under the robustness curve is

$$\mathrm{nAUC} = \frac{1}{\rho_{\max}} \int_0^{\rho_{\max}} a(\rho)\, d\rho. \tag{5}$$

The normalization by $\rho_{\max}$ makes the statistic interpretable on the same $[0, 1]$ scale as normalized accuracy. Integration is performed separately for every trained seed; means, uncertainty estimates, and paired regime differences are computed only after the per-seed nAUC values have been obtained.

The primary analysis uses $\rho_{\max} = 0.09$, with $\rho_{\max} = 0.095$ reported as an interval-sensitivity analysis. The primary and sensitivity endpoints are defined from the common achieved-displacement support of ImageNet-100, the limiting dataset, and are applied to both datasets to preserve identical integration intervals. The separate wider-range CIFAR-100 endpoint used in Section 5 is defined from the CIFAR-100 common support. The endpoint-selection criteria are described in Section 4.8.

4. Experimental Setup

4.1. Datasets and evaluation splits

We conduct experiments on CIFAR-100 and ImageNet-100. For every dataset, split seed 0 is used to construct a deterministic permutation and mutually disjoint calibration, attack, and evaluation subsets.

For ImageNet-100, the validation set contains 5000 images. We use 256 images for attention-logit displacement calibration, 1280 images for adversarial parameter optimization, and the remaining 3464 images for accuracy evaluation.

For CIFAR-100, the test set contains 10000 images. We use 256 images for attention-logit displacement calibration, 1280 images for attack optimization, and the remaining 8464 images for accuracy evaluation.

The calibration, attack, and evaluation subsets have zero pairwise overlap.

4.2. Positional-encoding families

Let P denote the number of image patches and $N = P + 1$ the total sequence length including the `[CLS]` token. The convolutional patch map is flattened in row-major raster order and the classification token is then prepended. Thus, sequence position $p = 0$ denotes `[CLS]`, whereas positions $p = 1, \ldots, P$ denote image patches in row-major order. None of the four baseline implementations uses an axial, factorized, or otherwise explicitly two-dimensional positional construction; image geometry enters these mechanisms through the one-dimensional raster sequence.

We study four positional-encoding mechanisms:

- **Learned**: a trainable absolute positional table $E_{\text{pos}} \in \mathbb{R}^{1\times N\times d}$ is added to the token representations. Its entries, including the independent `[CLS]` entry at $p = 0$, are initialized from $\mathcal{N}(0, 0.02^2)$;

- **Sinusoidal**: a deterministic one-dimensional sinusoidal table is added to the token representations,

$$E_{\text{pos}}(p, 2k) = \sin\!\left(p\,10000^{-2k/d}\right), \tag{6}$$
$$E_{\text{pos}}(p, 2k+1) = \cos\!\left(p\,10000^{-2k/d}\right). \tag{7}$$

  Consequently, `[CLS]` receives the $p = 0$ vector $(0, 1, 0, 1, \ldots)$;

- **RoPE**: rotary positional encoding uses the same one-dimensional raster index. With head dimension $d_h = 64$, the clean phase is $\theta_{p,k} = p\,10000^{-2k/d_h}$ for $k = 0, \ldots, 31$. Each block stores sine and cosine caches of shape $1 \times 1 \times N \times 32$, broadcast across all heads. Rotation uses the half-split convention, pairing the first and second halves of each head vector. The $p = 0$ `[CLS]` rotation is therefore the identity;

- **ALiBi-style**: a bidirectional, symmetric linear bias is added to the attention logits,

$$B^{(h)}_{ij} = -m_h|i - j|, \qquad m_h = 2^{-8(h+1)/H}, \qquad h = 0, \ldots, H - 1, \tag{8}$$

  where $H = 12$ for ViT-B/16 and $H = 6$ for ViT-S/16. The same architecture-specific fixed slope vector is used in every block and distance is computed over the complete one-dimensional sequence. No special `[CLS]` rule is applied, so its distance to patch position j is j.

Equation (8) is a monotone geometric slope schedule, not the reference non-power-of-two `get_slopes(12)` construction. For brevity, the tables and figures use the label “ALiBi” to denote this precisely defined ALiBi-style implementation. For the strongest-slope head, $m_0 \approx 0.630$ in ViT-B/16 and $m_0 \approx 0.397$ in ViT-S/16; on ImageNet-100 the corresponding `[CLS]`-to-final-position biases are approximately -123.5 and -77.8. These values impose architecture-specific head priors toward early indexed raster positions, although the realized attention also depends on the content-dependent query–key logits.

4.3. Architecture and training

ViT-B/16 uses $d = 768$, 12 blocks, 12 heads, head dimension 64, MLP ratio 4, and dropout 0.1; ViT-S/16 uses $d = 384$, 12 blocks, 6 heads, and the same head dimension, MLP ratio, and dropout. Within each architecture, models differ only in positional encoding.

For ImageNet-100, images are evaluated at 224×224 with 16×16 patches, producing 196 patch tokens and sequence length $N = 197$ after adding `[CLS]`. CIFAR-100 is evaluated only in the ViT-B/16 cohort at native 32×32 resolution with 4×4 patches, producing 64 patch tokens and $N = 65$.

All 72 canonical checkpoints are trained from scratch for 300 epochs using AdamW with initial learning rate 3×10^{-4}, weight decay 0.1, cosine annealing, 20 warmup epochs, batch size 128, label smoothing 0.1, Mixup with $\alpha = 0.8$, and global gradient-norm clipping at 1.0. The ViT-B/16 checkpoints are trained with CUDA automatic mixed precision and dynamic loss scaling; the canonical ViT-S/16 checkpoints are trained in full FP32. An auxiliary 18-checkpoint ViT-S/16 AMP cohort (three families by six seeds) follows the same training recipe and is used only in the within-ViT-S/16 training-regime and AMP-matched cross-architecture comparisons. The optimizer contains all trainable model parameters in one group, without positional-parameter, bias, or normalization exclusions from weight decay. Thus, the Learned positional table and trainable `[CLS]` token receive weight decay, whereas the Sinusoidal table, RoPE phase construction, and ALiBi slope state are fixed buffers during training. Canonical attack engines use float32 inputs and float32 model parameters without autocast or gradient scaling. They explicitly select the MATH scaled-dot-product attention backend where the installed PyTorch exposes `torch.nn.attention` (as in the locked PyTorch 2.8 audit environment); older-PyTorch fallback paths are not asserted to force MATH. The later audit numerical-mode capture records `matmul_allow_tf32=true` and `cudnn_allow_tf32=true`; hence "FP32" here denotes tensor/model dtype and the absence of mixed-precision autocast, not strict exclusion of TF32 matrix-multiply kernels. The detailed environment lock was captured for the later audit runs, whereas historical canonical-cohort environment reconciliation is explicitly retained as pending in the provenance report. Attack/evaluation precision is therefore distinct from training precision, and historical numerical-mode claims are made at the engine-source level rather than as a contemporaneous environment lock.

Table 1: Clean classification accuracy on the fixed robustness evaluation splits. Values are mean ± sample standard deviation over six independently trained seeds. ImageNet-100 uses 3,464 images and CIFAR-100 uses 8,464 images.

PE family	ViT-B ImageNet-100	ViT-B CIFAR-100	ViT-S ImageNet-100
Learned	79.20 ± 0.49	67.88 ± 0.49	77.95 ± 0.41
Sinusoidal	81.51 ± 0.54	67.07 ± 0.52	78.86 ± 0.38
RoPE	84.28 ± 0.27	73.03 ± 0.16	82.64 ± 0.34
ALiBi	80.97 ± 0.45	67.51 ± 0.37	79.21 ± 0.28

4.4. Checkpoint cohort and clean performance

For every positional-encoding family in each architecture–dataset cohort, we analyse six independently trained models with seeds

$$\{42, 123, 456, 789, 1011, 1213\}.$$

The study therefore contains 48 ViT-B/16 checkpoints (4 families $\times 6$ seeds $\times 2$ datasets) and 24 ViT-S/16 ImageNet-100 checkpoints (4×6), for a total of 72.

Within each architecture, comparisons between perturbation regimes and families retain the same seed identities and checkpoints. Across architectures, the seed labels are aligned for descriptive contrasts, but are not treated as strict pairs because model dimensionality, initialization, and training trajectories differ. Ordinary model weights remain frozen throughout robustness evaluation; only the family-specific positional representation is perturbed.

Table 1 reports clean accuracy on the same fixed evaluation populations used to normalize the robustness curves. ImageNet-100 uses the common 3,464-image evaluation split for both architectures, and CIFAR-100 uses the 8,464-image evaluation split. The ViT-S/16 runs used a re-extracted physical copy of the ImageNet-100 validation directory; the audit input lock verifies that the ViT-B/16 and ViT-S/16 copies have identical 5,000-image sample-order SHA-256 and identical 256/1,280/3,464 split SHA-256, so this path difference does not change the evaluated population. Every table cell is generated from the public seed-level clean-accuracy records.

The Learned table is the only positional state directly subject to AdamW weight decay. This optimizer asymmetry is part of the complete as-trained implementation and was not isolated as a separate causal factor.

4.5. Random parameter-noise protocol

Random robustness uses Gaussian directions in the unique PE parameter space, normalized by the global RMS convention. For each model and draw, the same direction is retained across the native budget grid, producing a coherent parameter-space path rather than independently resampled points.

Ten random directions are evaluated per model in the main robustness experiment. For PE family f, training seed s, native noise budget b, and fixed random directions $d = 1, \ldots, 10$, the achieved displacement and normalized accuracy are first averaged at the same native budget:

$$\bar{\rho}_{fsb} = \frac{1}{10}\sum_{d=1}^{10} \rho_{fsbd}, \qquad \bar{a}_{fsb} = \frac{1}{10}\sum_{d=1}^{10} a_{fsbd}. \tag{9}$$

The budget-paired points $(\bar{\rho}_{fsb}, \bar{a}_{fsb})$ define one random-noise curve per trained seed. We do not integrate the ten draw-specific curves separately and do not construct a union grid over their distinct achieved-displacement coordinates.

4.6. Adversarial parameter perturbations

Adversarial PE perturbations are generated using projected gradient ascent on cross-entropy loss over the fixed attack subset. We use five independent random restarts and retain the restart with the largest attack loss.

The step size is set to a fixed fraction of the native RMS budget:

$$\alpha = 0.05\varepsilon.$$

Family-specific optimization lengths were first selected in a seed-42 development audit and then checked on the prespecified confirmation seed 123. For ViT-B/16 ImageNet-100, the retained historical harmonization runner performs matched N-to-$2N$ comparisons with identical five-restart seeds on the fixed 256/1280/3464 calibration/attack/evaluation split. The run ledger records Learned 400×5-to-800×5, Sinusoidal 400×5-to-800×5, RoPE 200×5-to-400×5, and ALiBi 100×5-to-200×5 confirmation runs, followed by a targeted ALiBi 200×5-to-400×5 run at native budget 0.02. These records document the historical schedule-lock decision used below. The initial seed-42 step-doubling development audit did not use an a priori numerical convergence threshold; the retained schedules were supported by an observed empirical separation between stable and extended configurations. The confirmation seed 123 and the subsequent canonical grids were fixed separately, so the empirical development criterion is not presented as a prespecified threshold. The public repository redistributes the harmonization runner and run ledger, but not the original row-level comparison JSONs or the v17/v18 closure file; we therefore do not claim public row-level reproduction of that historical schedule decision.

For CIFAR-100, the final attack configurations are:

- Learned: 400 steps and 5 restarts;
- Sinusoidal: 200 steps and 5 restarts;
- RoPE: 100 steps and 5 restarts;
- ALiBi: 100 steps and 5 restarts.

For ViT-B/16 ImageNet-100, the final configurations are:

- Learned: 400 steps and 5 restarts;
- Sinusoidal: 400 steps and 5 restarts;
- RoPE: 200 steps and 5 restarts;
- ALiBi: 200 steps and 5 restarts.

All six ViT-B/16 ImageNet-100 ALiBi attack curves were computed under this final 200×5 schedule.

For ViT-S/16 ImageNet-100, an architecture-specific convergence and confirmation workflow locked:

- Learned: 1600 steps and 5 restarts;
- Sinusoidal: 200 steps and 5 restarts;
- RoPE: 200 steps and 5 restarts;
- ALiBi: 200 steps and 5 restarts.

The 1600×5 Learned schedule is supported by targeted matched longer-horizon checks; no claim is made that 3200×5 or 6400×5 was executed.

The attack is constrained in the family-specific native PE parameter space using the global RMS norm. It is not directly constrained by ρ_{rel}.

4.7. Construction of the final displacement grid

The mapping from native parameter budget to achieved ρ_{rel} is family and seed dependent. A uniform native budget schedule can therefore leave long interpolation intervals in attention-logit displacement space, particularly close to the clean operating point. We construct the final family-specific measurement grids to provide denser coverage in these regions while preserving the same attack objective, parameter norm, optimization length, and number of restarts.

For Learned PE on both datasets, the final grid includes the targeted low-budget native RMS values

$$\{0, 0.00025, 0.0005, 0.00075, 0.001\}.$$

For CIFAR-100 RoPE, it includes

$$\{0, 0.005, 0.010, 0.015, 0.020\},$$

and for ImageNet-100 RoPE it includes

$$\{0, 0.0125, 0.0375\},$$

with the final 200-step, five-restart attack configuration. In addition, the CIFAR-100 transition-region grid contains native RMS budgets 0.00625 for Learned, 0.01125 for Sinusoidal, 0.225 for RoPE, and 0.0375 and 0.060 for ALiBi. These five budgets were selected from the seed-42 development scan before the final cohort was analysed and were subsequently measured for all six seeds using the locked family-specific schedules. A retained historical audit summary reports that a same-protocol H200 recheck of the seed-42 transition points differed from the earlier RTX Blackwell Server Edition values by at most 4.91×10^{-4} in normalized accuracy and 2.56×10^{-5} in achieved ρ_{rel}. The earlier raw Blackwell log is not redistributed, so this is an audit-recorded cross-hardware check rather than a publicly rerunnable paired comparison; only the homogeneous H200 values are used in the final six-seed transition grid.

The locked ViT-S/16 ImageNet-100 adversarial native-budget grids are

$$\begin{aligned}
\text{Learned:}\ & \{0, 0.003, 0.004, 0.0045, 0.005, 0.006, 0.008\},\\
\text{Sinusoidal:}\ & \{0, 0.005, 0.006, 0.007, 0.0075, 0.008, 0.0085, 0.009, 0.0095, 0.010\},\\
\text{RoPE:}\ & \{0, 0.010, 0.025, 0.050, 0.075, 0.100\},\\
\text{ALiBi:}\ & \{0, 0.001, 0.0025, 0.005, 0.010, 0.015, 0.025, 0.040\}.
\end{aligned}$$

Random-noise curves use ten fixed Gaussian directions per checkpoint and family, retained across each native-budget path. Adaptive threshold refinement, when present, changes only the nominal budget grid and does not change steps, restarts, objective, split, or checkpoint selection.

Validated intermediate points are merged with the canonical measurements before curve construction. One analysis script then generates seed-level curves, nAUC tables, and the main robustness figure from the same curve objects while recording SHA-256 hashes of all consumed result files.

4.8. Statistical analysis

The trained checkpoint is the unit of replication. Curve construction and integration are performed separately for each seed before across-seed summaries; the ten random-noise draws are repeated within-checkpoint measurements, aggregated at matched native budgets according to Eq. (9), not independent replicates. Seed 42 was used to develop the family-specific PGD schedules and native grids and is therefore not fully independent of those choices. It remains in the prespecified six-seed cohort, with a five-seed sensitivity reported separately: all 40 remaining gaps are positive, family-mean orderings are preserved on both datasets, and the exact Friedman test gives $Q = 13.56$ and $p = 1.38648 \times 10^{-4}$ for each dataset. The development audit validates the protocol but is not independent confirmatory evidence.

For each seed, family, and regime, measured points are ordered by achieved ρ_{rel} and the exact clean anchor $(0, 1)$ is included. Adversarial curves use the cumulative lower envelope

$$\widetilde{a}_{\text{adv}}(\rho_i) = \min_{j \le i} a_{\text{adv}}(\rho_j), \tag{10}$$

so a weaker optimized attack at a larger native budget cannot improve the strongest-observed curve within the locked grid. This one-sided transformation is appropriate to the adversarial estimand: at each canonical budget we retain the highest-loss restart and then the lower envelope across achieved displacement. Random-noise curves instead estimate the mean response to isotropically sampled directions and are averaged across draws rather than enveloped. Audit-only budgets do not alter this locked primary estimand. A no-envelope sensitivity analysis (Supplementary Section 5, Table **??**) preserves all 48 positive seed-level gaps and all within-seed family rank configurations; the largest family-mean change is CIFAR-100 Learned, $0.015369 \to 0.014591$.

The primary statistic is normalized area under the accuracy–displacement curve,

$$\mathrm{nAUC}_{s,f,r} = \frac{1}{\rho_{\max}} \int_0^{\rho_{\max}} a_{s,f,r}(\rho)\, d\rho, \tag{11}$$

with piecewise-linear interpolation and trapezoidal integration. Endpoints are chosen solely from common achieved-displacement support, independently of accuracy outcomes or family rankings. The ImageNet-100 protocol initially planned $\rho_{\mathrm{rel}} \in [0, 0.1]$, but final coverage required a support-driven reduction: the smallest maximum adversarial displacement is 0.095748 (Sinusoidal), versus 0.115553 for noise. We therefore use $\rho_{\max} = 0.09$ as the primary endpoint and 0.095 as a near-maximal common-support sensitivity endpoint, applying both also to CIFAR-100 for cross-dataset comparability. Every included curve has measured support beyond the applicable endpoint; no nAUC uses extrapolation.

The within-seed decoupling statistic is

$$G_{s,f} = \mathrm{nAUC}_{s,f,\mathrm{noise}} - \mathrm{nAUC}_{s,f,\mathrm{adv}}\,. \tag{12}$$

We report its mean and sample standard deviation over six seeds and paired bootstrap confidence intervals obtained by resampling seed identities while preserving regime pairing. Family-specific exact two-sided sign-flip tests enumerate all 2^6 assignments under sign symmetry. Thus $p = 0.03125$ is the smallest attainable two-sided value when all six gaps share a sign; these eight tests are directional-consistency checks, not separate confirmatory claims (Bonferroni threshold 0.00625; Holm-adjusted values 0.25).

Between-family heterogeneity is assessed separately for each dataset by an exact Friedman permutation test on the four family gaps within each of the six shared seed labels. Shared labels provide blocks but do not make family-specific checkpoints strict training pairs because the parameterizations and optimization trajectories differ. With no ties, the exact null enumerates all $(4!)^6 = 191{,}102{,}976$ within-block rank assignments. We report Q and its exact permutation p; the asymptotic χ^2_3 value is reference only. The omnibus test addresses heterogeneity of gap magnitude, not positivity of the gap.

CIFAR-100 additionally permits a wider $\rho_{\max} = 0.23$ ordering-sensitivity analysis, below its smallest per-seed maximum adversarial displacement (0.236431). This post-primary interval is not pooled with the $[0, 0.09]$ inference and requires no extrapolation. No individual random draw, budget, layer, or image is treated as an independent replicate; standard deviations use Bessel's correction unless stated otherwise.

For cross-architecture ImageNet-100 analyses, support is again intersected over all included curves. The four-family canonical comparison uses $\rho_{\text{canonical}} = 0.0792383973$ and contrasts AMP-trained ViT-B/16 with full-FP32 ViT-S/16, so it does not isolate architecture from precision. The separate three-family AMP-matched comparison uses $\rho_{\text{cross}} = 0.0873455716$; each analysis also uses 0.9 and 0.75 of its own endpoint for sensitivity, without mixing support or extrapolating. Near-duplicate achieved-displacement points are deterministically grouped within absolute/relative tolerances 5×10^{-7} and 5×10^{-6}, respectively; ties retain lower normalized accuracy, then larger task loss, lower nominal budget, and stable source order.

Seed labels are aligned across architectures but are not matched training pairs: the backbones consume different amounts of the global random stream, and the DataLoader has no separately locked generator. We therefore report seed-aligned difference intervals together with unpaired Welch intervals as a sensitivity check. Normalized accuracy is not clipped at one. Finally, the all-restart Sinusoidal grid extension is explicitly post-lock, and the direct-displacement audit uses a different objective; neither changes the canonical support or primary nAUC estimand.

5. ViT-B/16 Random–Adversarial Robustness Decoupling

5.1. Primary robustness curves

Figure 1 shows the final merged robustness curves on the common achieved-displacement scale. The random-noise and adversarial responses separate to markedly different degrees across PE families and datasets, while the adversarial curves generally decline more rapidly over the primary range. The corresponding seed-aggregated nAUC values and paired gaps are summarized in Table 2.

5.2. Normalized area under the robustness curve

For each seed and perturbation regime, we integrate normalized accuracy over the common measured support and divide by the integration width to obtain nAUC. This support-matched scalar summarizes the full robustness curve while avoiding comparisons outside empirically covered ρ_{rel} values. Table 2 reports the resulting seed-level aggregates; the noise-minus-adversarial gap is interpreted primarily through its magnitude and family dependence rather than through positivity alone.

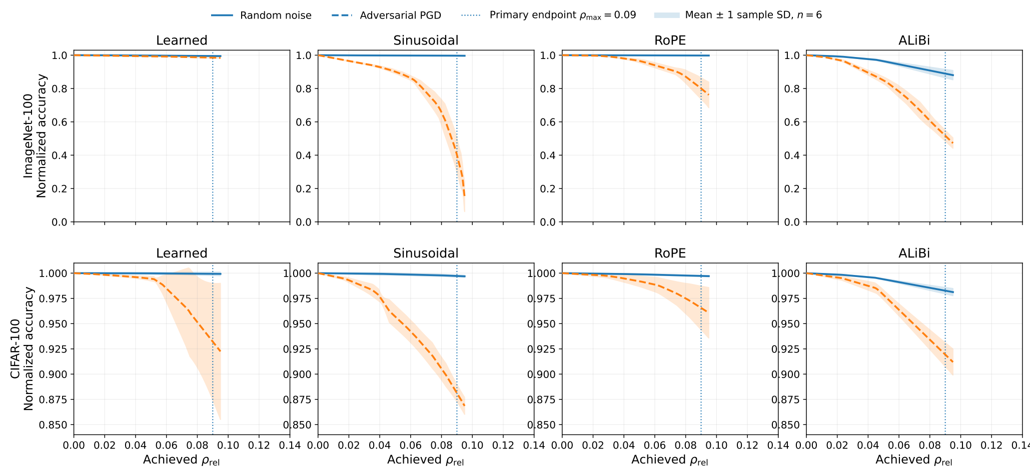

Figure 1: Normalized accuracy as a function of achieved relative attention-logit displacement for random parameter noise and adversarial positional-parameter perturbations. Solid and dashed curves show the across-seed mean, and shaded bands show mean $\pm$ one sample standard deviation over $n = 6$ independently trained seeds per PE family. The dotted vertical line marks the primary integration endpoint $\rho_{\max} = 0.09$. Curves are constructed and displayed over achieved attention-logit displacement rather than nominal native parameter budget. The two rows use dataset-specific y-axis limits to make the smaller CIFAR-100 deviations visible; each panel remains on the same normalized-accuracy scale within its dataset row. The random curves use the budget-paired ten-draw aggregation in Eq. (9); the figure and both nAUC tables are generated by the same canonical analysis script.

5.3. Family-dependent robustness gaps

Across both datasets, random-noise nAUC exceeds adversarial nAUC for every PE family. This direction is consistent across all six paired seeds in each dataset–family combination.

The result is not scientifically interesting merely because an optimized attack is stronger than random perturbation. The relevant finding is that the size of the random–adversarial gap is strongly family dependent and changes across datasets.

On ImageNet-100, Sinusoidal and ALiBi exhibit much larger separations between random and adversarial robustness than Learned PE. RoPE occupies an intermediate position. On CIFAR-100, all four gaps are smaller within the primary interval, and their relative ordering differs from that observed on ImageNet-100.

The interaction between PE family and perturbation regime therefore indicates that random parameter-noise stability is not a uniform surrogate

Table 2: Primary normalized area under the robustness curve over $\rho_{\mathrm{rel}} \in [0, 0.09]$. Values are mean $\pm$ sample SD across six paired training seeds. Random-noise curves use the budget-paired ten-draw aggregation defined in Section 4.8. Confidence intervals are paired seed-level bootstrap intervals for the noise-minus-adversarial gap and are pointwise, not simultaneous. The eight exact two-sided sign-flip tests each yielded the minimum attainable unadjusted value $p = 0.03125$ at $n = 6$; they are secondary directional-consistency checks and do not support multiplicity-adjusted individual significance claims (see Section 4.8).

Dataset	PE family	Noise nAUC	Adversarial nAUC	Gap	95% CI	Seeds with gap > 0
ImageNet-100	Learned	0.9980 ± 0.0011	0.9935 ± 0.0025	0.0045 ± 0.0027	$[0.0027, 0.0066]$	6/6
	Sinusoidal	0.9983 ± 0.0012	0.8610 ± 0.0205	0.1373 ± 0.0211	$[0.1219, 0.1529]$	6/6
	RoPE	0.9990 ± 0.0005	0.9516 ± 0.0134	0.0474 ± 0.0137	$[0.0382, 0.0582]$	6/6
	ALiBi	0.9602 ± 0.0081	0.8336 ± 0.0206	0.1266 ± 0.0231	$[0.1074, 0.1390]$	6/6
CIFAR-100	Learned	0.9998 ± 0.0009	0.9844 ± 0.0125	0.0154 ± 0.0122	$[0.0072, 0.0244]$	6/6
	Sinusoidal	0.9989 ± 0.0010	0.9584 ± 0.0043	0.0405 ± 0.0044	$[0.0379, 0.0441]$	6/6
	RoPE	0.9988 ± 0.0005	0.9902 ± 0.0060	0.0087 ± 0.0060	$[0.0044, 0.0131]$	6/6
	ALiBi	0.9935 ± 0.0010	0.9733 ± 0.0046	0.0203 ± 0.0037	$[0.0175, 0.0229]$	6/6

for adversarial positional-parameter robustness.

The between-family heterogeneity of gap magnitude was confirmed by the exact Friedman permutation test within each dataset: $Q = 16.4$, exact $p = 8.04 \times 10^{-6}$ for ImageNet-100 and $Q = 15.0$, exact $p = 9.47 \times 10^{-5}$ for CIFAR-100. For reference, the corresponding three-degree-of-freedom asymptotic χ^2 approximations yielded $p = 9.39 \times 10^{-4}$ and $p = 1.82 \times 10^{-3}$, respectively. Both exact results remain significant after correction for the two dataset-level omnibus tests. These omnibus tests address heterogeneity of gap magnitude, which carries the primary interpretation. The uniformly positive 48/48 seed-level direction is reported as a consistency check, together with the paired effect sizes in Table 2.

These conclusions do not depend on the adversarial lower-envelope construction. In Supplementary Section 5 and Table **??**, all 48 gaps remain positive, all within-seed family rank configurations and positive-seed counts are preserved, and the exact Friedman results are unchanged.

5.4. Decomposition of the robustness gap

Writing $D_{\mathrm{noise}} = 1 - \mathrm{nAUC}_{\mathrm{noise}}$ and $D_{\mathrm{adv}} = 1 - \mathrm{nAUC}_{\mathrm{adv}}$ gives $G = D_{\mathrm{adv}} - D_{\mathrm{noise}}$. The decomposition shows that the gap is driven mainly by adversarial degradation in several cells because random-noise nAUC remains near one. At the primary endpoint, ImageNet-100 Sinusoidal has $D_{\mathrm{noise}} = 0.0017$ and $D_{\mathrm{adv}} = 0.1390$, ImageNet-100 RoPE has 0.0010 and 0.0484, and CIFAR-100 Learned has 0.0002 and 0.0156. Random-noise degradation is more material for ImageNet-100 ALiBi (0.0398 versus 0.1664) and, proportionally, ImageNet-100 Learned (0.0020 versus 0.0065). Thus the paired gap remains the primary contrast, while its numerical magnitude often reflects a near-flat random-noise branch combined with a substantially lower adversarial branch.

5.5. Baseline accuracy as an alternative explanation

We also examined whether baseline model quality could provide a common alternative explanation for the gap. For this analysis, clean accuracy is taken from the same robustness-evaluation subset used to normalize each seed-level curve. On CIFAR-100, the four family means are perfectly inversely ranked (Spearman $\rho = -1.000$); across all 24 seed-level observations, $\rho = -0.774$ ($p = 9.25 \times 10^{-6}$), and after centering both variables within family the association is $\rho = -0.395$ ($p = 0.056$). The same pattern does not reproduce on ImageNet-100: ViT-B/16 gives $\rho = +0.317$ ($p = 0.131$), while the full-FP32 ViT-S/16 cohort gives $\rho = -0.311$ ($p = 0.140$). Baseline accuracy can therefore covary with the gap within a particular cohort, especially CIFAR-100, but it does not provide a single dataset- and architecture-independent explanation for the observed cross-family robustness structure.

5.6. Endpoint sensitivity

Extending the integration endpoint from $\rho_{\max} = 0.09$ to $\rho_{\max} = 0.095$ did not alter the qualitative conclusions. All eight dataset–family combinations retained positive paired gaps for all six seeds. The corresponding unadjusted descriptive two-sided sign-flip result remained $p = 0.03125$ in every case. The noise, adversarial, and gap rankings of the four PE families were unchanged on both datasets. At the extended endpoint, the mean gap ranged from 0.0048 for ImageNet-100 Learned PE to 0.1668 for ImageNet-100 Sinusoidal PE, and from 0.0100 for CIFAR-100 RoPE to 0.0448 for CIFAR-100 Sinusoidal PE. Because the within-seed family rankings were preserved, the exact Friedman permutation results reported in the preceding subsection were unchanged; the asymptotic reference results were unchanged as well. Full endpoint-specific seed-level and aggregate estimates are retained in the reproducibility package.

5.7. Architectural class and random-noise stability are insufficient predictors

The observed family ordering is not explained by a coarse classification of positional encodings according to where they enter the computation graph. Learned and Sinusoidal encodings both act through additive token-space representations, yet their adversarial robustness gaps differ substantially

Table 3: Wider-range CIFAR-100 ordering-sensitivity analysis over $\rho_{\text{rel}} \in [0, 0.23]$, the conservative common-support endpoint defined in Section 4.8. Values are mean $\pm$ sample SD of nAUC across the six training seeds. Random-noise curves use the same budget-paired ten-draw aggregation as the primary analysis. Both regimes are generated by the locked canonical analysis pipeline from the final merged measurement grid, including the completed six-seed transition points, without extrapolation. Per-seed values and input SHA-256 hashes are provided in the reproducibility package.

PE family	Random-noise nAUC	Adversarial nAUC
Learned	0.9976 ± 0.0025	0.7106 ± 0.0523
Sinusoidal	0.9922 ± 0.0008	0.7581 ± 0.0390
RoPE	0.9948 ± 0.0009	0.7912 ± 0.1205
ALiBi	0.9641 ± 0.0062	0.8475 ± 0.0146

on ImageNet-100. Likewise, RoPE and ALiBi both modify the attention computation rather than the input token embeddings, but they do not exhibit a common adversarial robustness profile.

The broad distinction between embedding-space and attention-space positional mechanisms is therefore insufficient to determine empirical adversarial positional robustness within the examined cohort. Random-noise stability is also insufficient: the family-specific noise ordering does not consistently preserve the adversarial ordering, and the extreme ranks of the two orderings reverse over the wider CIFAR-100 displacement interval.

These failures indicate that vulnerability cannot be reduced to the location of the positional operation or to average stability under isotropic perturbations. A more informative analysis must separate how efficiently particular parameter-space directions produce attention-logit displacement from how harmful the resulting displacement is for the task. The local directional probe in the following section examines these two factors separately.

5.8. Wider-range rank reversal on CIFAR-100

The difference between the two perturbation regimes becomes especially clear when the CIFAR-100 analysis is extended beyond the primary small-displacement interval to $\rho_{\text{rel}} \in [0, 0.23]$, the conservative common-support endpoint defined in Section 4.8. Table 3 reports the corresponding seed-aggregated nAUC values computed from the final merged measurement grid without extrapolation.

The extreme ranks reverse consistently across all six seeds: Learned PE attains a higher wider-range nAUC than ALiBi under random noise in six

Table 4: Three complementary comparisons used to separate system-level, training-regime, and architecture contrasts.

Analysis	Architecture contrast	Training contrast	Shared PE families	Inferential role
Canonical	ViT-B vs ViT-S	AMP vs full FP32	4	System-level replication; factors confounded
Within-ViT-S regime	ViT-S fixed	AMP vs full FP32	3	Within-backbone training-regime comparison
AMP-matched architecture	ViT-B vs ViT-S	AMP fixed	3	Cross-architecture effect with training regime fixed

of six seeds, whereas ALiBi attains a higher nAUC than Learned PE under adversarial perturbations in six of six seeds. At the family-mean level, Learned PE has the highest random-noise nAUC and the lowest adversarial nAUC, while ALiBi shows the opposite mean ordering.

The ordering of the two intermediate families is less stable: RoPE exceeds Sinusoidal in mean adversarial nAUC, but in only two of six individual seeds. RoPE also exhibits by far the largest adversarial seed variability in the cohort (sample SD 0.1205). We therefore do not interpret the complete four-family adversarial ranking as a stable result; the reversal claim concerns the extremes.

This extremal rank reversal provides a stronger form of evidence than a difference in robustness magnitude alone: the choice of perturbation regime changes the qualitative conclusion about which PE family is the most, and which the least, robust at the family-mean level. Consistent with the treatment of the family-specific sign-flip tests in Section 4.8, the 6/6 pairwise reversal counts are reported descriptively rather than as multiplicity-adjusted tests.

6. Factorial Architecture–Precision Decomposition and Objective-Specific Audits

The three cross-cohort contrasts answer different questions and are interpreted jointly rather than as independent robustness tests. Table 4 makes explicit which factor is held fixed in each comparison.

6.1. Canonical four-family cross-cohort comparison

This comparison uses the canonical ViT-B/16 AMP-trained and ViT-S/16 full-FP32-trained cohorts. It is informative about transfer between the two complete canonical systems, but architecture and training precision jointly vary and are not assigned separate causal interpretations. The ImageNet-100 comparison is integrated only over the measured common-support interval $\rho_{\text{rel}} \in [0, 0.0792383973]$. This endpoint is set by the smallest canonical support among all 96 architecture–family–seed–regime curves, namely ViT-S/16 Sinusoidal seed 456. Before the later all-restart audit, three targeted

Table 5: Cross-architecture ImageNet-100 nAUC over the locked common-support interval $\rho_{\mathrm{rel}} \in [0, 0.07924]$. Values are mean $\pm$ sample SD across six seeds.

Architecture	PE family	Noise nAUC	Adversarial nAUC	Gap
ViT-B/16	Learned	0.9983 ± 0.0010	0.9945 ± 0.0023	0.0039 ± 0.0024
	Sinusoidal	0.9984 ± 0.0010	0.9034 ± 0.0105	0.0950 ± 0.0110
	RoPE	0.9992 ± 0.0005	0.9669 ± 0.0083	0.0323 ± 0.0086
	ALiBi	0.9685 ± 0.0061	0.8701 ± 0.0185	0.0983 ± 0.0201
ViT-S/16	Learned	0.9981 ± 0.0007	0.8509 ± 0.0374	0.1471 ± 0.0374
	Sinusoidal	0.9988 ± 0.0011	0.7664 ± 0.0385	0.2324 ± 0.0390
	RoPE	0.9992 ± 0.0008	0.8662 ± 0.0397	0.1330 ± 0.0396
	ALiBi	0.9717 ± 0.0070	0.9046 ± 0.0064	0.0670 ± 0.0066

non-canonical ViT-S/16 Sinusoidal coverage-extension runs (seeds 123, 456, and 1213; native budgets through 0.020) attempted to reach $\rho_{\mathrm{rel}} = 0.09$ but were registered as `COMPLETE_TARGET_NOT_REACHED`. Their ledger "combined max" values, 0.088095530, 0.079238397, and 0.087695707, respectively, equal the corresponding canonical maxima to the precision recorded in the ledger; the extension therefore did not enlarge support for those seeds. The original row-level coverage-extension JSONs are not redistributed, so this statement is limited to the registered ledger and canonical source records. The primary estimand retains the prespecified canonical native-budget grids. Post-lock all-restart audit points and direct-displacement points are excluded, even when their achieved displacement falls inside the integration interval. Table 5 and Fig. 2 summarize this support-matched comparison; Supplementary Section 6 and Table **??** quantify the later ViT-S/16 Sinusoidal budget-0.020 sensitivity.

At both scales, RoPE has the highest and ALiBi the lowest mean random-noise nAUC. The two middle families are not cleanly separated: Sinusoidal exceeds Learned by only 0.000112 for ViT-B/16 and is higher in 3/6 seeds; for ViT-S/16 the mean difference is 0.000706 and is positive in 5/6 seeds. We therefore do not interpret their middle ordering as a robust distinction. The adversarial ordering, however, changes from Learned > RoPE > Sinusoidal > ALiBi for ViT-B/16 to ALiBi > RoPE > Learned > Sinusoidal for ViT-S/16.

A separate later all-restart audit, distinct from the historical coverage-extension runs above, provides public row-level results at native budget 0.020 for all six ViT-S/16 Sinusoidal checkpoints. A post-lock sensitivity adds its loss-selected 0.020 point to each seed curve while retaining the same cross-

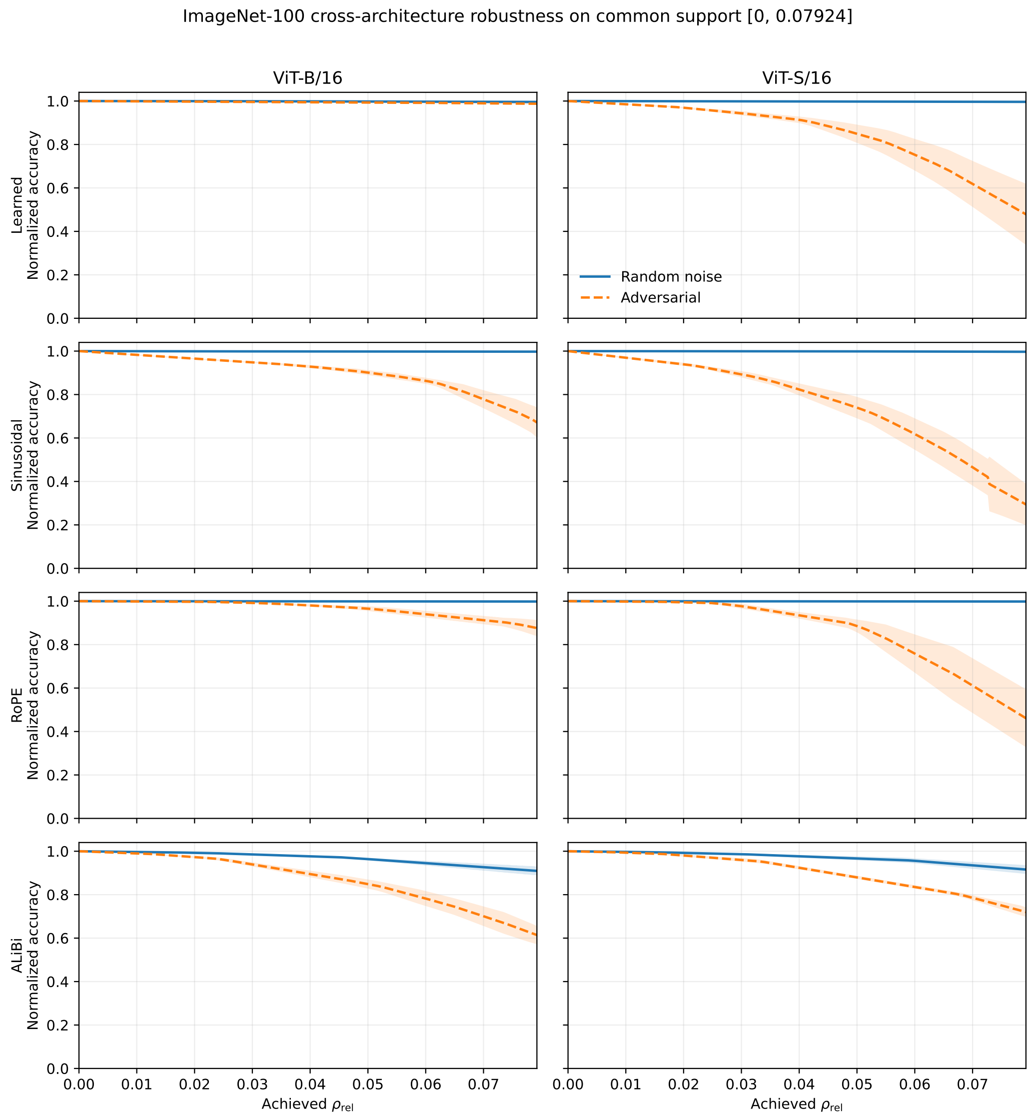

Figure 2: ImageNet-100 canonical robustness curves over the architecture-common support. Lines show across-seed means and shaded regions show ± 1 sample SD. Random noise is solid and task-loss adversarial perturbation is dashed. Every curve is constructed at seed level before aggregation and uses the prespecified canonical native-budget grid. Post-lock all-restart audit points and direct-displacement points are not included.

architecture endpoint. Its mean adversarial nAUC decreases from 0.766366 to 0.721777 ($\Delta = -0.044589$), increasing rather than weakening the random–adversarial gap; the qualitative family ordering is unchanged (Supplementary Table **??**).

For all eight combinations of architecture and family, the noise-minus-adversarial gap is positive in all six seeds. Exact Friedman permutation tests show between-family heterogeneity of gap magnitude for ViT-B/16 ($Q = 17.0$, $p = 2.386 \times 10^{-6}$) and ViT-S/16 ($Q = 15.0$, $p = 9.469 \times 10^{-5}$). The same family ordering and omnibus statistics are retained at 0.9 and 0.75 of the common endpoint. The cross-architecture mean differences are reported as seed-aligned descriptive contrasts, not strict paired effects.

6.2. Within-ViT-S training-regime comparison

We compared the completed ViT-S/16 cohorts trained with CUDA AMP–FP16 and full-FP32 using the same six seed labels and the same held-out split, attack budgets, parameter norm, restart count, and family-specific PGD schedules for Learned, Sinusoidal, and RoPE. Because bit-identical initialization hashes, training-sampler hashes, and per-run GradScaler skip counts were not packaged, we describe this as a *seed-matched*, rather than strictly paired, comparison.

The primary endpoint was $\rho(\varepsilon_{50})$, obtained by interpolating ρ through the same ε bracket in which the monotone accuracy envelope crossed 50% of clean accuracy. Before calculating cohort differences, we fixed an absolute equivalence margin of ± 0.01 in ρ_{rel}; equivalence required the two-sided 90% seed-matched confidence interval for the mean AMP-minus-FP32 difference to lie fully inside this margin.

Sinusoidal satisfied this criterion (AMP 0.06745 ± 0.00246 versus FP32 0.06797 ± 0.00508; $\Delta = -0.00052$, 90% CI $[-0.00498, 0.00394]$). Learned ($\Delta = 0.03447$, 90% CI $[-0.02455, 0.09350]$) and RoPE ($\Delta = 0.00647$, 90% CI $[-0.00448, 0.01742]$) were inconclusive. The aggregate ordering was nevertheless identical in both cohorts, Learned > RoPE > Sinusoidal for $\rho(\varepsilon_{50})$. Support-matched attack and random-noise nAUC rankings were also preserved at $\rho_{\max} = 0.0792384$ and at the 0.9 and 0.75 sensitivity endpoints. Thus the data support descriptive ranking stability and a family-specific Sinusoidal equivalence result, but not a general claim that robustness transfers unchanged between AMP and full-FP32 training.

Table 6: AMP-matched ImageNet-100 adversarial nAUC over $\rho_{\mathrm{rel}} \in [0, 0.08735]$. Architecture values are mean $\pm$ sample SD across six seeds; Δ is ViT-S minus ViT-B. Exact sign-flip values are secondary directional-consistency checks.

PE family	ViT-B adversarial nAUC	ViT-S adversarial nAUC	Δ adversarial	Seed-aligned 95% CI	Seeds $\Delta < 0$	Exact p	Δ noise nAUC
Learned	0.9937 ± 0.0024	0.7751 ± 0.0717	-0.2186	$[-0.2929, -0.1443]$	6/6	0.03125	-0.000894
Sinusoidal	0.8736 ± 0.0181	0.7137 ± 0.0198	-0.1599	$[-0.1842, -0.1356]$	6/6	0.03125	$+0.000729$
RoPE	0.9559 ± 0.0119	0.8627 ± 0.0557	-0.0932	$[-0.1604, -0.0261]$	5/6	0.06250	$+0.000132$

6.3. AMP-matched cross-architecture comparison

The within-ViT-S/16 comparison shows what changes when the training regime varies at fixed architecture. The AMP-matched comparison instead holds the training regime fixed and varies the backbone. The symmetric comparison contains Learned, Sinusoidal, and RoPE for the same six seed labels; ALiBi is excluded because ViT-S/16 AMP–FP16 training did not yield a valid checkpoint cohort. For both architectures, attack and evaluation use float32 tensors/model parameters without autocast or gradient scaling; the audit numerical-mode capture permits TF32 matrix multiplication as described above. The support intersection over both architectures, all three families, all six seeds, and both perturbation regimes is $\rho_{\mathrm{cross}} = 0.0873456$. Table 6 reports the corresponding seed-level architecture contrasts.

With six nonzero seed contrasts, attainable two-sided exact sign-flip p-values are quantized in steps of $2/64 = 0.03125$. Thus 0.03125 is the floor and 0.0625 the adjacent rung, not a finely resolved distinction between the families. Per-family p-values are therefore not treated as primary evidence; effect sizes, intervals, seed signs, and the omnibus interaction carry the interpretation. Unpaired Welch sensitivity intervals are nearly identical to the seed-aligned intervals and remain below zero for all three families (Supplementary Table **??**).

As shown in Fig. 3, the adversarial ranking changes from Learned > RoPE > Sinusoidal in ViT-B/16 to RoPE > Learned > Sinusoidal in ViT-S/16. Letting Δ denote ViT-S minus ViT-B adversarial nAUC, 5/6 seeds share the ordinal pattern Learned < Sinusoidal < RoPE; seed 42 is the sole exception. The corresponding rank sums are 8, 11, and 17, and the exact Friedman test gives $Q = 7.0$, $p = 0.02894$.

The interaction is also sign-consistent across pairwise architecture-effect contrasts. Learned minus RoPE has 95% CI $[-0.2353, -0.0155]$ with the same negative sign in 5/6 seeds; Sinusoidal minus RoPE has CI $[-0.1333, -0.00014]$ and is negative in 6/6; and Learned minus Sinusoidal has CI $[-0.1416, 0.0242]$ and is negative in 5/6. The architecture effect is therefore ordinally robust

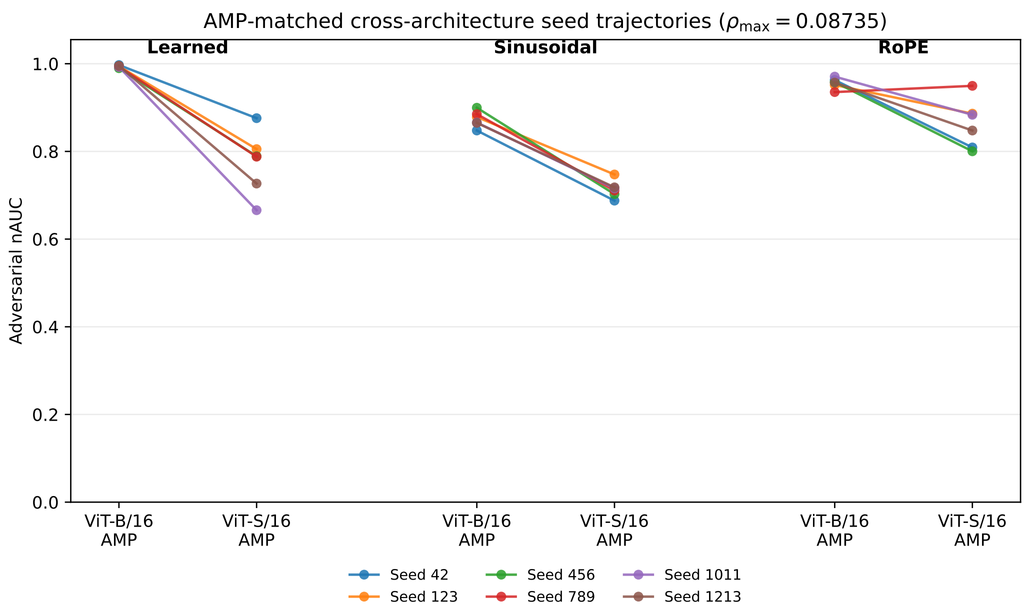

Figure 3: AMP-matched cross-architecture adversarial nAUC over $\rho_{\text{rel}} \in [0, 0.08735]$. Each line connects the same seed label between the ViT-B/16 AMP and ViT-S/16 AMP cohorts for one PE family. The labels are seed-aligned descriptive blocks rather than strict training pairs because backbone dimensionality, initialization streams, and optimization trajectories differ. Learned and Sinusoidal decrease in 6/6 aligned seeds, whereas RoPE decreases in 5/6.

across all three contrasts rather than resting on one pair.

Backbone scale also changes seed dispersion in a PE-dependent way. The ViT-S/ViT-B adversarial-nAUC SD ratios are 29.41 for Learned, 4.66 for RoPE, and 1.10 for Sinusoidal. For Learned, omitting any one seed leaves the SD ratio between 19.93 and 47.27, so the dispersion contrast is not attributable to one isolated run. Its ordering, Learned $\gg$ RoPE $>$ Sinusoidal, differs from the mean-effect ordering, Learned $>$ Sinusoidal $>$ RoPE. We report this as a descriptive second PE-dependent architecture effect without assigning a mechanism.

The random-noise control has no common seed-level direction: noise nAUC is lower in ViT-S for 3/6 Learned, 2/6 Sinusoidal, and 3/6 RoPE seeds, compared with 6/6, 6/6, and 5/6 negative adversarial contrasts. One ViT-S Sinusoidal noise nAUC is 1.00094 because normalized accuracy is not clipped at one and the noisy model marginally exceeds its clean reference on the finite evaluation split.

Threshold results remain consistent with the nAUC analysis. For RoPE, mean ϵ_{50} is essentially unchanged (0.067108 versus 0.067075), but mean ρ_{50} decreases from 0.116608 in ViT-B/16 to 0.083618 in ViT-S/16. For Sinusoidal, both ϵ_{50} and ρ_{50} decrease. The ViT-B/16 Learned curve does not bracket 50% normalized accuracy in three seeds and does not bracket 20% in any seed; consequently, a full $n = 6$ Learned threshold contrast is not reported. This censoring does not affect the measured-support nAUC analysis.

These results cover two backbone sizes and seed-aligned, not bit-identically paired, training runs. They establish that the mapping from achieved attention-logit displacement to task damage differs by PE family and backbone in the tested cohorts, not a universal scaling law or a strict causal estimate of model size.

6.4. The Sinusoidal task-loss frontier is objective-specific

The all-restart audit records all five Sinusoidal task-loss restarts at 48 combinations of architecture, seed, and budget, completing all 240 planned restarts. Under the task-loss objective, larger native budget can increase loss and reduce accuracy while the loss-selected global displacement stagnates or decreases. We therefore call the observed boundary a *Sinusoidal task-loss frontier*; it is not a structural upper bound nor an estimate of the largest attainable displacement.

A separate direct-displacement audit on ViT-S/16 FP32 Sinusoidal seed 42 shows that this frontier is objective-specific. Table 7 uses displacement

Table 7: Same-seed, same-budget comparison for ViT-S/16 FP32 Sinusoidal seed 42. Reported ρ_{rel} values are measured on the common 256-image calibration split; accuracies use the 3,464-image evaluation split.

Budget	Objective	ρ_{rel}	Accuracy (%)
0.0095	direct-ρ	0.10209	76.70
0.0095	task loss	0.09380	7.36
0.0200	direct-ρ	0.29505	19.52
0.0200	task loss	0.07945	1.70

from the same 256-image calibration split and accuracy from the same fixed 3464-image evaluation split for both objectives. At budget 0.0095, direct-displacement optimization reaches slightly larger global displacement yet preserves 76.70% accuracy, compared with 7.36% under task-loss optimization. At budget 0.020, direct-ρ reaches more than three times the task-loss displacement yet preserves much more accuracy. Global ρ is therefore not a sufficient scalar predictor of functional damage, and its relationship with damage is nonlinear in budget and direction.

The direct-ρ schedule is selected by a prespecified best-of-five criterion. Its best-of-five envelope is stable between 200×5 and 400×5, but individual restarts do not all converge. At budget 0.020, for example, paired restart 2 changes normalized accuracy from 0.66898 to 0.60212 between 200 and 400 steps, whereas the schedule-selected envelope changes by only 0.000365. The reported ρ is consequently an empirical lower bound on attainable displacement under the tested protocol, not a supremum estimate or evidence that the search was exhausted.

6.5. Layerwise displacement allocation

As shown in Fig. 4, direct-ρ perturbations are substantially more concentrated across layers than task-loss perturbations. The effective number of layers is 4.04 versus 9.22 at budget 0.0095 and 4.45 versus 10.98 at budget 0.020. Direct-ρ profiles are dominated by layer 2, whereas the task-loss profiles peak at layer 4; their profile cosines are 0.589 and 0.613. This supports direction-dependent inter-layer redistribution but does not identify particular heads or tokens and does not establish a causal mechanism.

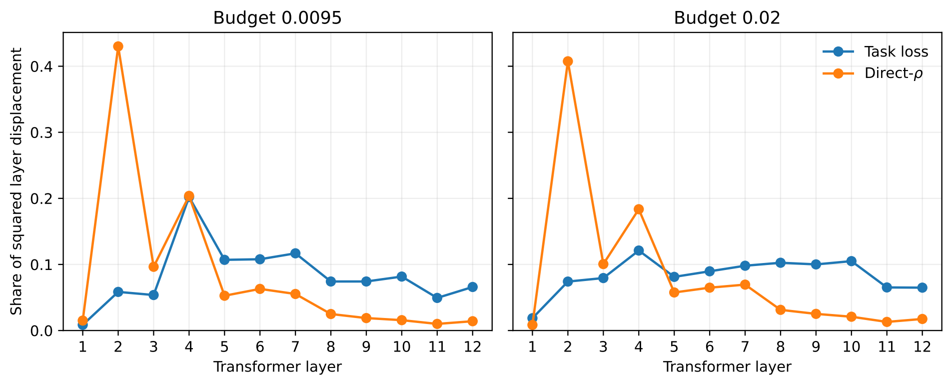

Figure 4: Layerwise share of squared attention-logit displacement for ViT-S/16 FP32 Sinusoidal seed 42. Direct-ρ and task-loss profiles are shown at the same nominal budgets.

7. Local Directional Geometry Probe

7.1. Method

The AUC analysis establishes whether random and adversarial robustness differ, but it does not determine why. We therefore perform a local directional probe on the 48 canonical ViT-B/16 checkpoints (two datasets $\times$ four PE families $\times$ six seeds). The ViT-S/16 cohorts are not probed in this analysis.

A task-gradient direction is estimated from a fixed subset $\mathcal{G}$ of 256 attack-split images:

$$g_p = \nabla_p \frac{1}{|\mathcal{G}|} \sum_{(x,y)\in\mathcal{G}} \mathcal{L}\big(f(x;\theta,p),y\big). \tag{13}$$

The normalized ascent direction is

$$u_{\text{task}} = \frac{g_p}{\|g_p\|_{\text{RMS}}}. \tag{14}$$

For comparison, we sample 32 Gaussian directions $u_{\text{rand}}^{(j)}$ and normalize each using the same global RMS convention.

For each dataset, the base split is generated once from the deterministic permutation with split seed 0. Denoting the resulting 256-image calibration split by $\mathcal{C}$ and the 1,280-image attack split by $\mathcal{A}$, the task-gradient subset is a fixed 256-image set $\mathcal{G} \subset \mathcal{A}$, whereas the geometry subset is a fixed 64-image set $\mathcal{H} \subset \mathcal{C}$. Consequently, $\mathcal{G} \cap \mathcal{H} = \varnothing$ by construction. The same dataset-specific indices are reused across all PE families and training seeds, preserving paired comparisons.

For each direction, its scalar magnitude is calibrated toward the nominal target

$$\rho_{\text{rel}} = 0.03$$

using the first 32 images of $\mathcal{H}$, denoted $\mathcal{H}_\rho \subset \mathcal{H}$. The calibrated direction is then evaluated on the complete 64-image set $\mathcal{H}$. Accordingly, all reported displacement, cross-entropy change, and damage-efficiency quantities use measurements on $\mathcal{H}$ rather than only on the inner calibration subset $\mathcal{H}_\rho$.

The nominal target is used only to place the directions in a common local neighbourhood. All final normalized quantities use the displacement actually measured over the complete set $\mathcal{H}$.

7.2. Directional functional amplification

For a parameter-space direction u, we define its local functional gain at probe radius r_0 as

$$\gamma(u) = \frac{\rho_{\text{rel}}(r_0 u)}{r_0}. \tag{15}$$

A fixed value $r_0 = 10^{-4}$ is used for every dataset, PE family, training seed, and probe direction. The radius is expressed in the same global-RMS-normalized native parameter-space units used to define the unit direction u. The locked value is recorded in `analysis/local_geometry/reference_outputs/probe_radius_evidence.json`, and the public direction-level table contains an explicit `probe_radius` column for all 1,584 direction records.

The task-to-random amplification ratio is

$$R_\gamma = \frac{\gamma(u_{\text{task}})}{\operatorname{median}_j \gamma(u_{\text{rand}}^{(j)})}. \tag{16}$$

A large value of R_γ indicates that the task-gradient direction changes the attention logits much more efficiently than the median of the 32 sampled Gaussian reference directions at the same parameter-space RMS magnitude. Because γ is a finite-radius local gain evaluated at $r_0 = 10^{-4}$, the numerical value of R_γ is radius dependent. The family ordering is preserved when gains are recomputed as secants at the calibrated $\rho_{\text{rel}} \approx 0.03$ operating point, but the magnitudes are not scale invariant.

7.3. Task-selective damage efficiency

Let α_u denote the calibrated scalar for direction u, and let

$$\rho_{\mathcal{H}}(u) = \rho_{\mathrm{rel}}(\alpha_u u; \mathcal{H})$$

be its actually achieved displacement on the complete geometry subset.

The corresponding increase in cross-entropy is

$$\Delta\mathrm{CE}(u) = \mathrm{CE}(p + \alpha_u u; \mathcal{H}) - \mathrm{CE}(p; \mathcal{H}). \tag{17}$$

We define damage efficiency as

$$\eta(u) = \frac{\Delta\mathrm{CE}(u)}{\rho_{\mathcal{H}}(u)}. \tag{18}$$

The task-selective damage gap is

$$\Delta\eta = \eta(u_{\mathrm{task}}) - \mathrm{median}_j\, \eta(u_{\mathrm{rand}}^{(j)}). \tag{19}$$

Because Eq. (18) uses the actually measured full-subset displacement, it does not assume that every calibrated direction achieves exactly the nominal target value of 0.03.

7.4. Scope of the local probe

The probe is a local directional diagnostic. Here, "held out" means separate from the images used to estimate the task-gradient direction: $\mathcal{H}$ comes from the fixed calibration partition and contains $\mathcal{H}_\rho$, so it is not an additional independent test set. The analysis does not recover globally dominant Jacobian directions; it instead compares task-aligned and random parameter directions while separating parameter-to-attention amplification from classification damage.

7.5. Directional amplification results

The task-gradient direction exhibits strongly family-dependent functional amplification. Values below are reported as mean $\pm$ sample standard deviation across the six independently trained ViT-B/16 checkpoints within each family.

On ImageNet-100, the task-to-random functional-gain ratio is $11.40\pm3.49\times$ for Learned PE, compared with $1.54\pm0.08\times$ for Sinusoidal PE, $1.25\pm0.03\times$ for RoPE, and $1.06 \pm 0.03\times$ for ALiBi. On CIFAR-100, the corresponding ratios are $5.81 \pm 0.33\times$, $3.42 \pm 0.14\times$, $1.69 \pm 0.42\times$, and $1.05 \pm 0.09\times$, respectively.

These results show that the local mapping from PE parameter space to attention-logit space is highly anisotropic for Learned PE and, to a lesser extent, CIFAR-100 Sinusoidal PE. In contrast, ALiBi exhibits task and random functional gains of similar magnitude. Functional gain, however, quantifies only how efficiently a direction reaches a displaced attention state; it does not quantify how harmful that state is for classification.

7.6. Task-selective damage results

Geometric amplification and task-selective damage are distinct. The mean task-minus-random damage-efficiency gap is therefore reported together with its sample standard deviation and the number of seeds in which the gap is positive. A positive family mean is not treated as seed-consistent evidence unless the seed signs support that interpretation.

On ImageNet-100, Learned PE combines the strongest amplification, $11.40 \pm 3.49\times$, with a much smaller damage-efficiency gap of 0.91 ± 1.10, positive in 6/6 seeds. Sinusoidal PE shows the complementary pattern: its amplification is only $1.54 \pm 0.08\times$, yet its damage-efficiency gap is the largest, 6.39 ± 2.91, and is positive in 6/6 seeds. RoPE has a near-zero mean gap of 0.09 ± 0.33, positive in only 4/6 seeds; this does not provide stable family-specific evidence of additional task-selective harmfulness. ALiBi has a positive mean gap of 2.25 ± 3.46, positive in 5/6 seeds, but the large between-seed variation requires a heterogeneous rather than uniform interpretation.

The same separation is especially clear on CIFAR-100. Learned PE has a large task-to-random gain of $5.81 \pm 0.33\times$, whereas its damage-efficiency gap is only 0.14 ± 0.33 and positive in 3/6 seeds. It therefore provides a direct counterexample to the idea that strong parameter-to-attention amplification necessarily entails additional task-selective damage. Sinusoidal, RoPE, and ALiBi show smaller amplification factors but more consistently positive gaps: 0.48 ± 0.13 in 6/6 seeds, 0.47 ± 0.59 in 5/6 seeds, and 0.63 ± 0.50 in 5/6 seeds, respectively.

The local geometry probe therefore distinguishes two measurable components:

1. how efficiently a parameter direction produces an internal attention-logit displacement;
2. how damaging that internal displacement is for the classification task.

Neither quantity alone is sufficient to explain positional-parameter robustness.

7.7. Summary of the geometry analysis

The probe separates directional amplification from task-selective harmfulness: Learned PE can strongly amplify the task-gradient direction without the largest damage per achieved displacement, whereas ImageNet-100 Sinusoidal shows moderate amplification but strongly task-selective damage. These results provide a mechanism-oriented interpretation, not an independent robustness ranking. They are local diagnostics based on one task-gradient direction and a finite random reference set, not a spectral characterization of the full Jacobian.

8. Discussion

8.1. Random robustness is not an adversarial-robustness proxy

Across both ViT scales, random-noise nAUC exceeds adversarial nAUC in every family and seed. The result is not merely that optimized attacks are stronger. Gap magnitude and family ordering vary with positional mechanism, dataset, and architecture, so stability under typical Gaussian directions does not identify the family most stable under task-aligned directions.

The canonical cross-cohort replication sharpens this conclusion. The random-noise family-mean extremes remain similar across the AMP-trained ViT-B/16 and full-FP32 ViT-S/16 cohorts, but the small intermediate family differences are not treated as a stable ordering; adversarial ordering changes substantially. The Learned–ALiBi extremes reverse: Learned is strongest and ALiBi weakest adversarially for ViT-B/16, but ALiBi is strongest and Learned falls below RoPE for ViT-S/16. This is a descriptive cross-cohort effect in which architecture and training precision jointly vary, not an isolated architecture effect or a universal scaling law.

The three contrasts form a factorial decomposition. The canonical comparison shows system-level transfer when architecture and precision jointly vary; the within-ViT-S/16 comparison holds architecture fixed and varies training regime; and the AMP-matched comparison holds AMP training fixed and varies architecture. In the AMP-matched comparison, ViT-S/16 adversarial nAUC is lower for all three shared families while random-noise changes have no common seed-level direction. The architecture effect is ordinally consistent across pairwise family contrasts, reverses the Learned–RoPE ordering, and changes seed dispersion most strongly for Learned. Thus PE family governs both the mean and the variability of how backbone scale maps achieved attention-logit displacement into task damage.

8.2. Equal displacement magnitude does not imply equal damage

The achieved ρ_{rel} controls aggregate RMS magnitude of attention-logit change, not orientation. Perturbations with equal global displacement can differ in allocation across layers and heads, query–key structure, token pattern, and alignment with task-critical directions.

The local geometry probe already separates functional amplification from task-selective harmfulness. ImageNet-100 Sinusoidal PE has only $1.54 \pm 0.08\times$ task-to-random functional gain but the largest damage-efficiency gap, 6.39 ± 2.91, positive in all six ViT-B/16 seeds. CIFAR-100 Learned PE provides the converse: despite $5.81 \pm 0.33\times$ amplification, its damage-efficiency gap is only 0.14 ± 0.33 and positive in 3/6 seeds.

The direct-displacement audit provides a stronger objective-level demonstration. For ViT-S/16 Sinusoidal seed 42, direct-ρ optimization at budget 0.0095 achieves slightly larger displacement yet preserves 76.70% accuracy, compared with 7.36% under task-loss optimization. At budget 0.020, direct-ρ reaches much larger displacement but remains less destructive. The layerwise profile analysis shows that these directions also distribute displacement differently across blocks. Magnitude, orientation, and task damage are therefore separate quantities.

8.3. Family-specific and architecture-specific geometry

The families differ in parameter dimension, sharing, and injection: Learned is a high-dimensional additive space, Sinusoidal changes deterministic positional state, RoPE is tied through phase rotations, and ALiBi uses low-dimensional slopes shared across blocks. Architecture changes therefore alter both the native perturbation space and its mapping into attention.

The strong early-raster prior of the studied ALiBi-style implementation may contribute to its distinctive behavior, but the experiments do not isolate that mechanism causally. Likewise, the direct-ρ layer profile is descriptive and cannot establish which heads, tokens, or downstream computations cause the observed accuracy changes.

8.4. Implications for robustness evaluation

Positional-robustness evaluation should report random and task-aligned perturbations separately, compare them by achieved pre-softmax logit displacement, and integrate only over measured common support. Post-lock points from the same objective belong to grid-extension sensitivity analyses; different-objective points remain separate. Accordingly, the all-restart audit tests restart/grid sensitivity, whereas direct-displacement optimization probes the objective-specific displacement–damage relationship.

With six contrasts, exact two-sided sign-flip p-values are coarse (0.03125 is the attainable floor). Seed-level effects, magnitude, dispersion, interval agreement, ordinal consistency, and the omnibus interaction are therefore more informative than fine-grained interpretation of family-wise p-values.

8.5. Limitations

The canonical study covers two datasets for ViT-B/16 but only ImageNet-100 for ViT-S/16. Two architecture sizes cannot establish a general scaling law. Equal seed labels across architectures do not form strict training pairs because the backbones consume different random streams during initialization and do not guarantee identical data order; unpaired Welch sensitivity intervals nevertheless agree with the seed-aligned intervals. The canonical four-family comparison also confounds architecture with training precision; the AMP-matched comparison removes that confound for three families but cannot include ALiBi because the ViT-S/16 AMP–FP16 training recipe did not yield a valid checkpoint cohort. The public evidence for this exclusion is cohort-level governance; raw per-seed ALiBi training logs are not redistributed, so no exact per-seed failure count is claimed. Additional model sizes, patch resolutions, and datasets are required for broader generalization.

All four baseline constructions use one-dimensional row-major patch ordering. The conclusions do not establish the behavior of axial, factorized, or explicitly two-dimensional variants. AdamW weight decay also acts on the trainable Learned positional table but not deterministic buffers, so comparisons characterize the complete as-trained systems rather than a weight-decay-invariant causal effect of positional form.

Canonical adversarial perturbations are optimized under native family-specific parameter constraints and reparameterized afterward by achieved ρ_{rel}; they are not directly ρ-constrained. The family-specific schedule lock rests on an observed empirical separation in the seed-42 step-doubling development audit rather than an a priori numerical convergence threshold; the confirmation-seed check is therefore interpreted as validation of that empirical choice, not as evidence that the threshold itself was preregistered. The direct-displacement audit partially probes this distinction but is restricted to ViT-S/16 FP32 Sinusoidal seed 42. Its schedule-level PASS concerns the best-of-five envelope, not convergence of every restart or exhaustion of the search. No cross-seed or cross-architecture direct-ρ generalization is made.

Because ρ_{rel} is defined before softmax, it also retains row-constant logit shifts in the exact null space of row-wise softmax. The fraction of measured displacement in this null space can in principle depend on PE family and perturbation direction; cross-family rankings are therefore rankings on the stated pre-softmax logit scale rather than claims of equality under a fully softmax-invariant metric. A row-centered quotient-space sensitivity metric would be complementary, but is not part of the canonical protocol evaluated here.

The scalar displacement aggregates layers, heads, queries, and keys. The layerwise displacement-profile analysis resolves the layer dimension for one objective comparison but does not resolve heads or tokens and is not causal. The local geometry probe is restricted to the 48 ViT-B/16 checkpoints and is not evaluated on ViT-S/16; it uses one task-gradient direction and 32 Gaussian directions rather than a complete Jacobian spectrum. With six checkpoints per family, mixed-sign family-specific damage summaries remain descriptive.

9. Conclusion

We compared Learned, Sinusoidal, RoPE, and ALiBi under random parameter noise and task-loss adversarial perturbations on a common achieved attention-logit displacement scale. The canonical evidence spans 72 checkpoints: AMP-trained ViT-B/16 on CIFAR-100 and ImageNet-100 and a full-FP32 ViT-S/16 ImageNet-100 replication, with six seeds per family and cohort. A separate 18-checkpoint ViT-S/16 AMP cohort enables within-backbone precision bridging and AMP-matched cross-architecture analysis.

For ViT-B/16, every dataset–family–seed comparison has a positive noise-minus-adversarial nAUC gap over the primary interval $[0, 0.09]$. Over the architecture-common ImageNet-100 support $[0, 0.07924]$, the gap remains positive in all six seeds for every family at both model scales, with significant between-family heterogeneity. The random-noise extremes are stable across the canonical cohorts, but adversarial ordering is not: Learned is strongest for ViT-B/16, whereas ALiBi is strongest for the full-FP32 ViT-S/16 cohort and Sinusoidal becomes weakest. In the AMP-matched three-family comparison, all ViT-S/16 adversarial nAUC means are lower than their ViT-B/16 counterparts, random nAUC changes have no common seed-level direction, and Learned and RoPE reverse order. The architecture effect is ordinally consistent in 5/6 seeds and is accompanied by strongly PE-dependent seed dispersion.

The local geometry probe, all-restart audit, direct-displacement audit, and layerwise displacement profiles jointly show why a single scalar robustness label is inadequate. Parameter-to-attention amplification, global displacement, inter-layer allocation, and classification damage can vary independently. The Sinusoidal boundary observed under task-loss optimization is therefore reported explicitly as a task-loss frontier, and direct-ρ points are kept outside canonical support.

Random-noise stability is not a reliable proxy for empirical adversarial positional-parameter robustness. Positional-encoding families differ in how achieved attention-logit displacement is converted into functional damage, and that mapping changes with backbone architecture. Evaluation should therefore combine random and task-aligned directions, achieved attention-logit displacement, measured common-support integration, seed-level dispersion, and explicit objective and architecture qualifiers.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The public reproducibility repository `vit-pe-decoupling` contains the manuscript source, analysis code, immutable input hashes, seed-level and aggregate numerical outputs, figures, protocol and split manifests, the local directional-geometry records, the clean-accuracy anchors, and the all-restart, direct-displacement, layerwise, no-envelope, and post-lock-grid sensitivity records used in this article. Trained checkpoint binaries are not duplicated in the manuscript archive. For the 48 canonical ViT-B/16 checkpoints, the historical attack JSONs record checkpoint paths rather than contemporaneous digests; the complete SHA-256 manifest was reconstructed post hoc from the retained checkpoint cohort, and all 48 attack paths bind exactly to it. Six ImageNet-100 Sinusoidal hashes also match independently retained historical digests. ViT-S/16 full-FP32 and AMP checkpoint manifests are likewise distributed. The public audit bundle also includes the later numerical-mode lock, its engine-source provenance report, and the Sinusoidal audit input lock that verifies identical ImageNet-100 sample ordering/split hashes across the original ViT-B/16 validation copy and the re-extracted ViT-S/16 copy. These later audit records are not presented as contemporaneous environment captures for the historical canonical runs. The archived reproducibility package for release v1.0.0 is available on Zenodo at `doi:10.5281/zenodo.21979816`.

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